An AI Literacy Curriculum

An AI Literacy Curriculum Built on the Lunar Olivine Frontier

The Moon is my beautiful impossibility.

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Tutorial • Colab Notebooks • Lunar Petrographic Exhibit

This curriculum is built around the planetary science grand challenge: *How can AI assist scientists in using thin sections from returned lunar samples to reconstruct the Moon's volcanic history, crystallization processes, and mantle evolution?* Recognizing that future scientific breakthroughs will increasingly depend on the ability to integrate vast and complex datasets, synthesize knowledge across disciplines, and leverage AI as a tool for discovery, this curriculum immerse learners in scientific investigations that combine planetary science, AI and data-driven inquiry. Through the analysis of lunar thin sections and the study of olivine-rich lunar samples, learners critically assess the capabilities and limitations of AI systems while exploring how AI can assist scientific observation, pattern recognition, and evidence-based reasoning in reconstructing the Moon's formation and evolution.

Contents

Learning Outcomes

By the end of the curriculum, learners will be able to:

- Identify olivine in lunar thin sections and clearly distinguish between observations, AI-generated predictions, and scientific interpretations when analyzing evidence.
- Assess the strengths, limitations, and potential sources of error or bias in AI systems used for scientific investigations.
- Curate ML-ready datasets from lunar thin section images and lunar olivine data for AI-assisted analysis.
- Design and develop lunar thin section microscopy exhibits and lunar olivine exhibits.
- Advanced learners will develop AI-assisted workflows for analyzing olivine-bearing lunar samples and AI-assisted exhibits.

Audience and Prerequisites

This curriculum is designed for advanced secondary students, lower-division undergrads, and broader public audiences. No prior geology background is required. No prior knowledge of machine learning is assumed either. This curriculum can be implemented as either an intensive multi-day workshop or an extended learning experience spanning several weeks.